# RES=RAG and the Coherent State Network Protocol

A Formal Architecture for Endogenous Stability in Human-Machine Systems

Canonical formalization and deterministic reference profile, version 1.0.0

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**Epistemic status.** This is a formal specification and testable research programme. It does not report a completed empirical validation, prove machine consciousness, or claim that human-machine interaction is a physical quantum system.



**Abstract**

RES=RAG models stable intelligence as a regulated tension between receptive, context-grounded dynamics (RES) and internally generative, novelty-producing dynamics (RAG). The central claim is not that the two distributions become equal. Stability instead occupies a calibrated, non-zero optimal-transport band: too little separation produces rigidity, while excessive separation produces instability, fabrication, or loss of coordination.

This paper consolidates the conceptual framework introduced by Jean-Charles Tassan; the computational formalization and systems-validation work developed by Trent Slade; and the CSNP, dialogical-time, mathematical-structure, and endogenous-stability lineages developed by Mohamad Al-Zawahreh, Bertrand D. J.-F. Thébault, Manuel Martín Morales Plaza, Wilson John Sterking Lauret, Dominique Colin, Fatiha (Nisrine) Bouzid, and Timothy Sullivan. It supplies a common state space, measurable quantities, explicit failure conditions, and a protocol boundary. The Coherent State Network Protocol (CSNP) turns the model into a cyclic observation-classification-intervention-receipt process. A deterministic reference profile, CSNP-RP 1.0.0, defines canonical JSON receipts, hash chaining, calibration declarations, evidence references, and replay rules.

The result is deliberately modest in epistemic scope. RES=RAG is presented as a falsifiable architecture for studying regulated dialogue and distributed human-machine coordination. Thresholds are calibration parameters, not universal constants; Wasserstein distances depend on declared encoders and sampling procedures. The contribution is therefore a reproducible theory and protocol specification from which empirical studies can be designed.

## 1 Scope and provenance

RES=RAG begins from a relational thesis: stable cognition is not adequately described by reception or generation alone, but by their regulated co-presence across human, machine, and interaction domains. The originating framework and three-axis organization are due to Tassan. Subsequent joint work introduced an optimal-transport reading, a history-dependent informational substrate, dual operators, and interface-level observables [Tassan and Slade, 2025, Slade and Tassan, 2025, 2026]. The dialogical-time, mathematical-structure, CSNP/Wasserstein, and endogenous-stability antecedents were developed across the wider team [Tassan and Thébault, 2025, Tassan et al., 2025, Al-Zawahreh et al., 2026, Lauret et al., 2026]. The canonical repository then separated axioms, predictions, falsifiers, and ethical constraints [Tassan et al., 2026].

This version has four goals:

1. preserve the canonical distinction between receptive and generative dynamics;
2. express equilibrium as a bounded, non-zero transport relation;
3. connect the theory to observations available at a human-machine interface; and
4. define an auditable protocol and deterministic receipt format.

Only work directly connected to RES=RAG is included. Broader field theories, unrelated QSOL-IMC systems, and uncited external projects are outside scope.

### 1.1 Claim labels

| Label | Meaning in this paper |
|---|---|
| **Definition** | A convention that gives the specification a precise meaning. |
| **Protocol rule** | A normative requirement for CSNP-RP conformance. |
| **Hypothesis** | A claim that can be tested and rejected. |
| **Reported target** | A value from exploratory work that is not established here. |
| **Limitation** | A known boundary on interpretation or implementation. |

Table 1: Epistemic labels used throughout the formalization.

Claims such as a 40-fold memory reduction, hallucination below 0.02 percent, coherence near 0.96, or universal collapse thresholds are not results of this formalization. If adopted by an implementation, they must be marked as external, unvalidated targets and coupled to a citable measurement procedure.

## 2 Formal objects

Let $(\mathcal{S}, d)$ be a declared Polish metric space, and let $\mathcal{P}_2(\mathcal{S})$ be the probability measures on $\mathcal{S}$ with finite second moment. A state can represent a semantic embedding, dialogue state, task state, or another reproducible encoding. The representation, sampling rule, and ground metric are part of the study specification.

**Definition 1** (Functional distributions)**.** *At observation $t$,*

$$\mu_t^{\mathrm{RES}} \in \mathcal{P}_2(\mathcal{S}) \quad \textit{and} \quad \mu_t^{\mathrm{RAG}} \in \mathcal{P}_2(\mathcal{S})$$

*denote respectively the receptive, contextual, evidence-conditioned distribution and the generative, novelty-producing distribution.*

These are functional roles. RAG here means regenerative or generative dynamics; it is not restricted to the engineering pattern usually called retrieval-augmented generation.

For $\mu, \nu \in \mathcal{P}_2(\mathcal{S})$, the quadratic Wasserstein distance is

$$W_2^2(\mu, \nu) = \inf_{\gamma \in \Pi(\mu,\nu)} \int_{\mathcal{S}\times\mathcal{S}} d(x, y)^2 \, \mathrm{d}\gamma(x, y), \tag{1}$$

where $\Pi(\mu, \nu)$ is the set of couplings with marginals $\mu$ and $\nu$ [Villani, 2009, Ambrosio et al., 2008]. Define

$$w_t = W_2\left(\mu_t^{\mathrm{RES}}, \mu_t^{\mathrm{RAG}}\right). \tag{2}$$

## 2.1 Bounded equilibrium

**Definition 2** (RES=RAG equilibrium). *For calibration profile $\theta = (\varepsilon_{\min}, \varepsilon_{\max}, \ldots)$, where $0 < \varepsilon_{\min} < \varepsilon_{\max}$, the stable set is*

$$\mathcal{E}_\theta = \left\{(\mu^{\mathrm{RES}}, \mu^{\mathrm{RAG}}) : \varepsilon_{\min} \le W_2(\mu^{\mathrm{RES}}, \mu^{\mathrm{RAG}}) \le \varepsilon_{\max}\right\}. \tag{3}$$

The equality sign in RES=RAG names this regulated relation. It does not require pointwise identity or zero transport. Below the lower boundary, the process risks sterility, repetition, or over-constraint; above the upper boundary, it risks drift, fabrication, or loss of shared reference.

The distance-to-band diagnostic

$$e_\theta(w_t) = \max(0, \varepsilon_{\min} - w_t) + \max(0, w_t - \varepsilon_{\max}) \tag{4}$$

is non-negative and vanishes precisely inside the declared stable interval.

**Proposition 1** (Non-minimization). *Minimizing $w_t$ without the lower bound is not a valid RES=RAG objective.*

*Proof.* An unconstrained minimum admits $w_t = 0$. By equation (3), that point is outside the stable set whenever $\varepsilon_{\min} > 0$, and it erases the productive difference the model is intended to preserve. □

## 2.2 Temporal-reference discrepancy

Let $\mu_t^H$ and $\mu_t^M$ be declared human-side and machine-side state estimates.

**Definition 3** (Operational real-time discrepancy).

$$T_{\mathrm{Real}}(t) = W_2(\mu_t^H, \mu_t^M). \tag{5}$$

$T_{\mathrm{Real}}$ is a discrepancy between situated reference frames. It is not physical clock time. A conforming empirical study must declare the estimator, window, encoder, ground metric, and uncertainty. Its provenance includes the dialogical-time work of Tassan and Thébault [Tassan and Thébault, 2025].

## 2.3 History-dependent information state

Let $s_t \in \mathcal{M}$ be a history-dependent information state. The informational-substrate formalization motivates the phenomenological dynamics

$$\frac{\mathrm{d}s}{\mathrm{d}t} = \mathcal{G}(s, t) - \mathcal{R}(s, t), \tag{6}$$

where $\mathcal{G}$ is a generative operator and $\mathcal{R}$ is a receptive-regulatory operator [Slade and Tassan, 2025]. These operators are not physical forces. A study using equation (6) must specify how their outputs induce the probability measures in equation (2); operator balance alone does not prove membership in $\mathcal{E}_\theta$.

# 3 Axes and interpretive modes

The framework separates three coupled domains.

| Axis | Domain | Example observation |
|---|---|---|
| **I** | Human internal regulation | Attention, affective stability, stated uncertainty. |
| **II** | Human-machine relational regulation | Semantic alignment, repair, pacing, and shared reference. |
| **III** | Machine functional regulation | Response distribution, retrieval grounding, novelty, and constraint density. |

Table 2: The three RES=RAG axes. Axis II is a relational process, not an attribute of either participant alone.

Two interpretive modes constrain inference:

- **Mode A, phenomenological or anthropomorphic:** language that treats the machine as if it possessed a first-person interior.
- **Mode B, functional or instrumental:** language tied to observable behavior, declared state, and reproducible measurement.

Mode B is the implementation and safety default. Mode A can be studied as a user-facing phenomenon, but must not be converted into evidence of machine consciousness. This discipline also reduces anthropomorphic and relational risk [Bender et al., 2021, Weidinger et al., 2022].

# 4 Interface-level observables

The framework permits mesoscopic measurement without privileged access to model internals [Slade and Tassan, 2026]. Let $P_t(z)$ be a declared or reconstructed distribution over candidate next units $z$. Its entropy is $H(P_t)$ [Shannon, 1948, Cover and Thomas, 2006]. For a window of $T$ observations, define normalized constraint density

$$\rho_c(t) = \frac{1}{T} \sum_{i=t-T+1}^{t} \left(1 - \frac{H(P_i)}{H_{\max,i}}\right). \tag{7}$$

The normalization $H_{\max,i}$ must be declared. If token probabilities are unavailable, a proxy receives a distinct metric identifier.

For observation interval $\Delta\tau_t > 0$, define propagation rate

$$v_c(t) = \frac{\rho_c(t) - \rho_c(t-1)}{\Delta\tau_t}. \tag{8}$$

Let $\kappa_t > 0$ estimate the response capacity of the observer or governance loop in the same reciprocal-time unit. With a declared numerical floor $\epsilon > 0$, define the dimensionless pressure

$$D_r(t) = \frac{\max(0, v_c(t))}{\max(\kappa_t, \epsilon)}. \tag{9}$$

**Hypothesis 1** (Rate-capacity warning)**.** *$D_r(t) > 1$ predicts a loss of governability before semantic failure.*

This is a warning condition, not proof of irreversible failure. Its accumulation/propagation behavior and endogenous-governability lineage are attributed to the wider team [Lauret et al., 2026].

# 5 Coherent State Network Protocol

In version 1.0.0, CSNP means *Coherent State Network Protocol*. Earlier work used *Client-Side Narrative Protocol* for a sovereign-memory implementation. The earlier term is preserved as an optional storage profile, not as the canonical expansion of the protocol name. This terminology history includes the work of Al-Zawahreh and the multi-author CSNP/Wasserstein synthesis [Al-Zawahreh et al., 2026].

## 5.1 Observed state

At step $t$, CSNP observes

$$X_t = (\delta_t, D_c(t), D_r(t), m(t), w_t, S_{\text{org}}(t), F(t)) , \tag{10}$$

where:

- $\delta_t = T_{\text{Real}}(t)$ is human-machine reference discrepancy;
- $D_c(t) = \rho_c(t) \in [0, 1]$ is normalized constraint density;
- $D_r(t) \geq 0$ is the rate pressure in equation (9);
- $m(t) \in [0, 1]$ is memory saturation under a declared estimator;
- $w_t \geq 0$ is the RES-RAG Wasserstein separation;
- $S_{\text{org}}(t) \in [0, 1]$ is a declared organizational incoherence score; and
- $F(t) \in [0, 1]$ is accountable anchoring availability.

An unimplemented metric is unavailable, not zero.

Given intervention $u_t$, an implementation realizes

$$\Phi_\theta(X_t, u_t) \longrightarrow (X_{t+1}, C_t, A_t, R_t) , \tag{11}$$

where $C_t$ is a classification, $A_t$ an admissible action, and $R_t$ a deterministic receipt.

## 5.2 Protocol cycle

1. **Declare**: select versioned metric and calibration profiles.
2. **Observe**: collect measurements and evidence references.
3. **Classify**: apply the profile with no hidden thresholds.
4. **Intervene**: select the least-force admissible action.
5. **Receipt**: canonicalize, hash, and chain the observation.
6. **Replay**: verify integrity, evidence, rules, and outcome.
7. **Recalibrate**: change thresholds only in a new profile version.

## 5.3 Classifications

**Governable** $w_t$ is within its band, $D_r \leq 1$, memory and organizational measures are below warnings, and anchoring is sufficient.

**Critical** At least one warning is exceeded, but an admissible intervention is expected to return the process to the stable set within the declared horizon.

**Irreversible** Under a declared finite intervention set and test horizon, no admissible intervention returns the observed process to the calibrated stable set. This is a protocol result, not a metaphysical claim.

**Indeterminate** Required measurements, evidence, or calibration data are missing or invalid.

**Protocol rule 1** (Irreversibility)**.** *A single threshold crossing must not produce an irreversible classification. The intervention set, evaluation horizon, uncertainty policy, and recovery criterion are required.*

### 5.4 Reference interventions

| Action | Intended effect |
|---|---|
| `none` | Continue observation inside the stable band. |
| `slowdown` | Reduce interaction or generation rate. |
| `diversify` | Restore productive novelty below the lower band. |
| `retrieve` | Add verifiable external grounding. |
| `clarify` | Repair ambiguity or conflicting state estimates. |
| `human_handoff` | Transfer authority to an accountable human. |
| `stop` | End an unsafe or unauditable process. |

Table 3: The CSNP-RP reference action vocabulary. Intervention success is measured, not assumed.

## 6 Deterministic reference profile

CSNP-RP 1.0.0 defines machine-readable receipts for equation (11). A receipt contains the protocol version, observation and calibration identifiers, normalized measurements, classification, intervention, evidence references, previous-receipt digest, state digest, and receipt digest.

### 6.1 Canonicalization

The JSON Canonicalization Scheme defined by RFC 8785 [Rundgren et al., 2020]:

1. sorts object keys lexicographically by UTF-16 code unit;
2. preserves array order;
3. serializes strings as well-formed UTF-8 JSON without insignificant whitespace;
4. serializes finite numbers with the exact ECMAScript representation required by JCS; and
5. preserves JSON boolean and null literals.

The state projection contains `metric_profile`, `measurements`, `observation_id`, `observed_at`, and `previous_receipt_hash`. Its canonical UTF-8 bytes are hashed with SHA-256 to obtain `state_hash`. The complete receipt, with only `receipt_hash` omitted, is then canonicalized and hashed to obtain `receipt_hash`.

For a chain of receipts $R_0, \ldots, R_n$,

$$R_t.\texttt{previous\_receipt\_hash} = \mathrm{SHA256}(\mathrm{canonical}(R_{t-1} \setminus \{\texttt{receipt\_hash}\}))\,. \tag{12}$$

The first link uses null. Integrity does not establish sensor truth, measurement validity, confidentiality, or actor trust.

### 6.2 Reference classifier

For complete measurements, the profile admits `governable` only if

$$\begin{aligned} \varepsilon_{\min} \le w_t &\le \varepsilon_{\max}, \\ D_r(t) &\le 1, \\ m(t) &< K_M, \\ S_{\mathrm{org}}(t) &< K_{\mathrm{org}}, \\ F(t) &\ge F_{\min}. \end{aligned} \tag{13}$$

The values $K_M$, $K_{\mathrm{org}}$, and $F_{\min}$ are versioned calibration parameters. They are not universal constants.

### 6.3 Replay boundary

A complete replay parses the receipt with strict duplicate-key and number handling, validates its schema, recomputes both digests, verifies the chain and evidence hashes, loads the exact metric profile, reruns the classifier, and records any disagreement. The bundled dependency-free verifier performs strict JSON parsing, structural, hash, and predecessor-link checks. Evidence retrieval and full JSON Schema validation remain integration responsibilities.

## 7 Safety constraints

RES=RAG yields design constraints rather than a single maximization objective:

1. do not optimize RES-RAG separation toward zero;
2. distinguish retrieved evidence from generated content;
3. retain estimator uncertainty and missing data;
4. do not infer machine consciousness from anthropomorphic language;
5. do not use user agreement as an Axis II stability proxy;
6. preserve accountable human control over consequential intervention;
7. classify unauditable state as indeterminate rather than safe; and
8. give any client-side narrative store export, deletion, provenance, consent, and access controls.

## 8 Hypotheses and falsification programme

Table 4: Testable RES=RAG and CSNP hypotheses.

| ID | Hypothesis | Example falsifier |
|---|---|---|
| **H1** | Task quality peaks within a calibrated non-zero $W_2$ band. | Quality is monotonic in distance or unaffected across preregistered tasks. |
| **H2** | $D_r > 1$ predicts governability loss before semantic failure. | It has no out-of-sample predictive value over matched baselines. |
| **H3** | Slowdown improves recovery under high rate pressure. | Randomized slowdown does not improve return-to-band time. |
| **H4** | Relevant retrieval reduces above-band drift. | Retrieval worsens or does not change controlled grounded accuracy. |
| **H5** | Diversity interventions improve below-band sterility. | Novelty-adjusted task quality does not improve. |
| **H6** | Axis II adds predictive value beyond separate Axis I and III measures. | A joint model does not outperform preregistered separate baselines. |
| **H7** | Hash-chained receipts improve incident reconstruction. | Blinded teams reconstruct incidents equally well without receipts. |
| **H8** | Style-only politeness is insufficient to recover high-density, high-rate states. | Style-only intervention matches rate- or density-targeted intervention. |

Each study should preregister the representation, encoder, ground metric, window, thresholds, missing-data rule, intervention set, outcome, and baseline. Thresholds fitted and tested on the same data are not independent validation.

## 9 Limitations

- Wasserstein values are representation-dependent; encoders and ground metrics can produce incompatible distances.
- Interface proxies do not identify internal mental or computational states without additional assumptions.
- The framework establishes neither human nor machine phenomenal consciousness.
- "Quantum-like" denotes contextual or non-classical behavioral structure only, not physical quantum computation.
- The dual-operator equation is phenomenological and may fail for discrete, path-dependent, or non-stationary systems.
- Deterministic receipts establish provenance integrity, not semantic truth.
- This release is a specification and verifier, not a benchmark, clinical instrument, or production safety certification.

## 10 Research pathway

A disciplined programme proceeds from synthetic traces to held-out studies:

1. construct traces with known drift and recovery events;

2. compare declared semantic representations and sensitivity to ground metrics;
3. calibrate thresholds on development data and freeze them;
4. test table 4 on held-out human-machine interactions;
5. publish receipts, metric profiles, code, uncertainty, and negative results; and
6. evaluate client-controlled memory under privacy, deletion, and adversarial-integrity tests.

Claims of broad societal impact should follow reproducible evidence. The immediate contribution is narrower: RES=RAG now has a coherent mathematical interpretation, an operational protocol, explicit falsifiers, and an auditable implementation boundary.

## Authorship and contributions

The conceptual origin of RES=RAG, its receptive-generative distinction, and its three-axis relational structure are attributed to Jean-Charles J. C. Tassan. The computational interpretation, optimal-transport architecture, history-dependent information model, operational variables, deterministic protocol profile, validation boundary, and release engineering are attributed to Trent Slade, developed in collaboration with Tassan. The CSNP, dialogical-time, mathematical-structure, and endogenous-stability provenance is attributed to the complete team: Mohamad Al-Zawahreh, Bertrand D. J.-F. Thébault, Manuel Martín Morales Plaza, Wilson John Sterking Lauret, Dominique Colin, Fatiha (Nisrine) Bouzid, and Timothy Sullivan, together with Tassan and Slade [Al-Zawahreh et al., 2026, Tassan and Thébault, 2025, Tassan et al., 2025, Lauret et al., 2026].

Tassan reviewed the consolidated work and confirmed on 27 July 2026 that all members of the team should be included in its citation. The repository therefore records all nine names as version 1.0.0 authors. The accompanying provenance statement distinguishes theory origin, formalization and software, and antecedent research lineages without implying that every author wrote every line or implemented every artifact. CRediT describes contribution rather than determining authorship [Brand et al., 2015]; more granular CRediT roles, affiliations, and ORCID identifiers may be added when confirmed by the individual team members.

## Artifact availability

The normative schema, protocol, verifier, tamper tests, Markdown manuscript, LaTeX source, bibliography, and compiled PDF are maintained in the public repository:

`https://github.com/QSOLKCB/res-rag`

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# A Normative artifact map

| **Path** | **Role** |
|---|---|
| `paper/RES_RAG_CSNP_Formalization_v1.0.0.tex` | Archival LaTeX source. |
| `paper/RES_RAG_CSNP_Formalization_v1.0.0.pdf` | Compiled review copy. |
| `paper/RES_RAG_CSNP_Formalization_v1.0.0.md` | GitHub-readable manuscript. |
| `paper/CONTRIBUTIONS.md` | Confirmed authorship and provenance record. |
| `protocol/CSNP-RP_v1.0.0.md` | Normative protocol rules. |
| `protocol/csnp-rp-v1.0.0.schema.json` | Receipt JSON Schema. |
| `protocol/examples/csnp-receipt.example.json` | Conforming synthetic receipt. |
| `protocol/tools/verify_receipt.mjs` | Dependency-free integrity verifier. |
| `protocol/tests/verify_receipt.test.mjs` | Validity and tamper tests. |

Table 5: Version 1.0.0 artifact set.